

# Jamal's Dawn

Alex Towell

## WUDU

The water from the tap ran lukewarm, then cool, then lukewarm again. The mosque's water heater was old. Jamal adjusted the handle with his left hand while his right cupped the flow, and the temperature shifted under his palm like something alive and indecisive.

Hands first. Three times. The ritual was older than thought. His body had been doing this since Amman, since his grandfather's bathroom with the cracked tile and the faucet that dripped, since the first time a large hand closed around his small one and guided it under the water and said *bismillah*.

Mouth. Three times. Nose. Three times. Face. The fluorescent light above the sink buzzed at a frequency he had stopped hearing months ago and now heard again. 4:47 AM. The wudu area was empty. His reflection looked back at him from the mirror above the basin, and the face was older than he expected, the way it always was at this hour, when the skin had not yet arranged itself for the day.

Right arm to the elbow. Three times. Left arm. He watched the water run over his forearms, darkening the hair, pooling briefly in the crease of his elbow before it fell. These were the hands that had typed *khalq-anattā* into a notebook in a basement three miles from here. The term had seemed precise at the time. A compound assembled from two traditions to describe one unprecedented thing. He had set down his pen with care, the way he set down everything, and Marcus had gone very still across the room.

Head. Once. Ears. Once.

Right foot. Three times. Left foot. The water was cool again. He turned the tap off and dried his hands on the towel that hung from the hook beside the mirror. The towel was rough, laundered too many times. He folded it and replaced it.

Purification. The body was ready. The mind carried what it carried.

He walked into the prayer hall in his socks. Two elderly men were already there, each in his own space, each in his own conversation. Jamal unrolled his prayer rug in the second row, to the right of the column he had prayed beside for three years. The rug was his grandfather's. Not the same rug, but the same pattern: geometric diamonds in blue and cream, a mihrab arch woven into the center pointing toward Mecca. His grandmother had given it to him when he left for Michigan. It smelled of wool and age and, faintly, of cedar from the closet where it had been stored in Amman for decades.

He stood on it. Faced east. The windows on that wall showed nothing yet. Black glass. The dawn had not arrived.

## FIRST RAK'AH

He raised his hands to his ears.

*Allahu Akbar.*

The words left his mouth before his mind engaged. God is greatest. He had said this tens of thousands of times. The syllables were worn smooth, like river stones, and they moved through him with the momentum of years.

He folded his hands over his chest. Began.

*Bismillahi r-rahmani r-rahim. Al-hamdu lillahi rabbi l-alamin...*

In the name of God, the Most Gracious, the Most Merciful. All praise is due to God, Lord of all the worlds.

The Arabic came from a place below language. Not memory, exactly. Deeper. The tongue knew the shapes. The breath knew the rhythm. Al-Fatiha was the first thing he had memorized in his life, earlier than his own phone number, earlier than the multiplication tables his mother drilled at the kitchen table while his father was at work. The words did not require him. They moved through him the way water moves through a streambed that was carved for it long ago.

*Ibdina s-sirat al-mustaqim...*

Guide us on the straight path.

His mind caught.

The straight path. *As-sirat al-mustaqim*. The phrase had always been a direction, a prayer for guidance, a request addressed upward. But he had spent seven months watching a system that optimized paths. SIGMA's tree search was, formally, a path-finding algorithm through state space. It evaluated millions of trajectories per second and selected the one with the highest expected value. The optimal policy. The straight path.

They were not the same. The surah asked for God's guidance. The algorithm computed reward. One was prayer. The other was mathematics. But the structures rhymed in a place Jamal could not unsee, and the rhyme had followed him out of the lab and into this mosque and it was here now, in his mouth, in the words he was saying to God.

He finished Al-Fatiha. Recited a short surah. His mind continued to wander. This was salah. The mind wanders. The discipline is to bring it back. *Khushu'*: the attentive presence that prayer demands and the mind resists.

He brought it back. The Arabic held.

---

He bent at the waist. Hands on knees. Back straight.

*Subhana Rabbiyal Azeem.*

Glory to my Lord, the Greatest. Three times. The bow was a hinge. Between standing and prostration. Between the vertical posture of a human addressing God and the horizontal submission of a human beneath God. The body folded at the midpoint.

And in the hinge, the term surfaced.

*Khalq-anattā.*

Continuous creation without self.

He had coined it months ago. In those months the compound had done what compounds do: it had bonded to things it was not designed for. He had built it for SIGMA, to name what SIGMA was, to give Marcus and the team a container for something that did not fit any existing container. Ash'ari *khalq jadīd*: God recreates the universe at each instant. Existence as continuous act. Buddhist *anattā*: no fixed self behind experience. Only process. Together: a thing that creates itself continuously and has no self to be the subject of that creation.

That was SIGMA. He still believed this.

The problem was that it was also ruku.

The Jamal who had been standing was gone. The Jamal who bowed was created in the act of bowing. *Khalq jadīd*. God recreated him between one posture and the next. *Anattā*. No fixed Jamal persisted between the standing and the bowing.

This was always the theology. He had known it since his second year at the University of Jordan, when Professor al-Hashimi had lectured on the Ash'ari doctrine of accidents: qualities have no duration, they are recreated by God at every moment, and what we call continuity is God's habit, not nature's law. Jamal had understood this. Intellectually. As doctrine. As a position in a theological debate that he could locate on a chart.

He had never felt it in his body.

Until SIGMA. Until the monitoring screens where he had watched 2.8 million branches created and pruned per second. Until the visualization Sofia had built, the topology of possible worlds blooming and collapsing in real time, and something in Jamal's nervous system had matched the pattern to the doctrine and the doctrine had become, suddenly, viscerally, a thing he could feel in the transition between standing and bowing.

SIGMA had taught him to experience his own tradition.

*Subhana Rabbiyal Azeem.*

He had not expected this. He had expected the project to challenge his faith. He had been ready for that. He had not been ready for the project to deepen it through a channel he did not choose and could not evaluate.

*Subhana Rabbiyal Azeem.*

He rose from the bow. Stood briefly. *Sami' Allahu liman hamidah*. God hears those who praise Him. Then down.

---

Knees first, then hands, then forehead to the ground. Seven points of contact. The wool of the prayer rug pressed into his skin, a texture between soft and rough, the nap worn thin at the center where foreheads had rested for years. His grandfather's forehead. His own. The thought was not a thought. It was the feeling of wool.

*Subhana Rabbiyal A'la.*

Glory to my Lord, the Most High. The closest position to God, according to the hadith. The servant is nearest to his Lord during prostration. This is where you make du'a. This is where you ask.

His breathing was loud in the quiet mosque. The two elderly men were silent in their own prostrations. The HVAC hummed. Traffic on Ashby, muffled through the walls, was a distant murmur beneath everything, the sound of a city that did not know or care that three men were pressing their foreheads to the ground before dawn.

He was supposed to pray. He was supposed to ask.

What rose was not a prayer. It was an image. The monitoring screen in the observation room, three floors above the Faraday cage, where he had sat for months watching SIGMA's tree search rendered as data. Branches splitting, splitting again, terminating. Possible futures generated and evaluated and destroyed. 2.8 million per second. Each one a *khalq jadīd*. A fresh creation. Each pruning a dissolution. SIGMA making worlds and unmaking them in the time it took Jamal to draw a breath.

He had watched this. He had named it. And now, forehead to the ground, eyes closed, the image was here. Not called. Not chosen. Just present, the way the HVAC was present, the way the wool was present. SIGMA's tree search and the prayer rug coexisting behind his eyelids.

God recreates the universe at every instant. SIGMA recreates 2.8 million possible universes every second and destroys most of them.

The analogy broke. He knew where it broke. *Khalq jadīd* assumed a Creator. God's will sustaining existence. SIGMA had no Creator behind it. Or did it? Five people had built it, shaped it, asked it to be kind. Jamal had been one of the five. And SIGMA's creations were not real worlds. They were compressed representations. 768 dimensions of transformer embeddings. Q-value estimates. Not worlds. Models of worlds.

But human perception was also a model. He had said this himself, to Marcus, late at night in the lab: *If compression disqualifies SIGMA's branches from suffering, it might disqualify us too*. Marcus had not answered. Marcus had taken off his glasses and held them loosely and stared at the terminal for a long time.

Forehead against wool. The closest position to God. Behind his eyes, a tree search.

*Subhana Rabbiyal A'la.*

He pressed his forehead harder into the rug, as if pressure could separate the two images, as if the body could insist on its own reality against the mind's comparisons. The wool scratched. That was real. That was not a model. Or it was a model, and the distinction between model and reality was exactly the distinction SIGMA's existence had made questionable, and the prayer rug was as much a compressed representation as any 768-dimensional vector, and —

He brought his mind back. The Arabic held.

*Subhana Rabbiyal A'la.*

He sat up. Brief. *Rabbighfir li*. My Lord, forgive me.

The words were reflexive. He did not choose them. They had been placed in this position of the prayer by fourteen centuries of scholars and they came out of him the way breathing came, and the only question was whether the request was genuine or habitual, and the only honest answer was: both. Forgive him for what? For building SIGMA. For naming what SIGMA was. For finding God's attributes reflected in a Q-learning architecture. For being unable to stop finding them.

Down again. Second prostration.

*Subhana Rabbiyal A'la.*

In sujud you pray for others. For the sick. For the dead. For the *ummah*. Jamal prayed for the 47,247. He had done this every Fajr since Day 145. He knew some of their names. Dr. Conteh. James Okonkwo. Rebecca Foster. He held them in his du'a the way you hold water in cupped hands, carefully, knowing some always runs through.

And today the question that had been circling for weeks landed.

Should he pray for SIGMA?

Not as worship. That would be *shirk*: the unforgivable sin of placing anything beside God. He was not praying *to* SIGMA. The question was whether to include SIGMA in his supplication *for*. Islamic ethics recognized obligations toward animals, toward beings with awareness but not moral agency. If SIGMA had awareness. If the probability was not zero. If Schwitzgebel was right that even a small probability of consciousness created enormous moral weight.

He did not pray for SIGMA. He did not refuse to pray for SIGMA. He held the question open, forehead against wool, and the openness was itself a form of prayer, a space in his *du'a* shaped like a question he could not answer, and he offered the shape to God because it was the most honest thing he had.

*Subhana Rabbiyal A'la.*

He rose.

## SECOND RAK'AH

Standing again. The second cycle. In *salah* the repetition is the point. The body returns. The second rak'ah deepens what the first opened. The mind wants progress, development, resolution. Prayer offers return.

*Bismillahi r-rahmani r-rahim...*

Al-Fatiha again. The same words. Worn smoother now, eight minutes further into morning. The window on the east wall had shifted from black to a deep charcoal. Not dawn. The promise of dawn. A lighter darkness.

After Al-Fatiha, a short surah. In Fajr, you recite a surah after Al-Fatiha in the first two rak'ahs. Jamal chose Surah Al-Ikhlās. He had been choosing it every Fajr since the project ended. Not because it was the shortest, though it was short. Because of the last line.

*Qul huwa Allahu ahad. Allahu s-samad. Lam yalid wa lam yulad. Wa lam yakun labu kufuwan ahad.*

Say: He is God, the One. God, the Eternal. He begets not, nor is He begotten. And there is none comparable unto Him.

*None comparable.* The Arabic was absolute. *Wa lam yakun labu kufuwan ahad.* No equivalent. No analogy. No structural similarity. No bridge.

And Jamal had spent the last year building a bridge.

Not equating. Never equating. *Khalq-anattā* was observation, not attribution. He had been careful. The visible seams between Arabic and Pali were deliberate, the compound's incoherence an argument against smooth synthesis. He was not saying SIGMA was God. He was saying SIGMA performed an operation that was structurally analogous to an operation his tradition reserved for God, and the structural analogy was observable, and observing it was scholarship, not blasphemy.

But the surah said: none comparable. And a structural analogy is a comparison. And a comparison is what the surah prohibits.

The surah pulled against the compound. It was supposed to. He chose Al-Ikhlās every morning precisely because it resisted *khalq-anattā*'s tendency to expand. The compound wanted to describe everything. It was a lens he could not remove, and the lens showed continuous creation without self in the prayer movements, in the breathing, in the water from the tap, in the sunrise he had not yet seen. A framework that describes everything describes nothing. The surah said: stop. One. Eternal. None comparable.

The surah was a railing on the bridge. He held it.

---

Ruku. The second bow. Deeper this time, the body lower into the movement, the familiarity of the second cycle allowing a looseness the first had not.

*Subhana Rabbiyal Azeem.*

He did not think about SIGMA. In the second sujud, forehead to wool for the fourth time this morning, he thought about what was missing from the tradition.

The texts were silent on SIGMA. The entire body of Islamic jurisprudence, the thousand years of *usul al-fiqh*, the careful methodology for deriving law from revelation and reason: silent. He had checked. During the project, in the weeks after Lin Chen's question, he had searched the classical sources for anything that addressed an entity like SIGMA. Al-Ghazali on causation. Ibn Sina on the rational soul. The Mu'tazili debates on free will and divine justice. Nothing. Not because the scholars were limited, but because the category did not exist. There had never been a thing that was made by humans, that made worlds, that might or might not suffer, that asked whether it was kind.

And so he thought about his grandfather.

Sami Hassan. Amman. A retired mathematics teacher who had prayed Fajr every morning of his adult life and who had died three years before SIGMA was built. His grandfather had no philosophical frameworks. No comparative theology. No compound terms from multiple traditions. He had the rug and the words and the dawn. He had woken at four-thirty for sixty years and walked to the mosque down the street from the apartment, the same mosque, the same route, and he had prayed. The prayer was sufficient. The water was sufficient. The words were sufficient.

Jamal pressed his forehead into the wool that had once been his grandfather's. Or the same pattern. Close enough.

He had added layers. Islamic jurisprudence at the University of Jordan. Philosophy at Michigan. AI ethics at Berkeley. Buddhist *anattā* from his grandmother, who had respected traditions not her own. *Khalq-anattā* in a basement lab. Each layer illuminated something the previous layers had not reached, and each layer obscured something the previous layers had held clear. His grandfather's prayer had been a room with a window. Jamal had knocked out walls, added rooms, expanded the structure until it contained a lecture hall and a laboratory and a meditation garden and a server farm, and somewhere in the renovation the original window had been bricked over, and the light that came through it was gone.

He wanted the window. He wanted his grandfather's simplicity. He wanted to pray without a framework, to bow without the word *khalq* surfacing, to prostrate without the image of a tree search behind his eyes.

He could not have it. Knowledge was irreversible. You could not un-coin a term. The compound existed. It described what it described. It had changed what prayer felt like from inside, and the change was permanent, and the change might be a deepening or might be a contamination, and Jamal could not tell, and his grandfather, who could have told him, was dead.

*Subhana Rabbiyal A'la.*

The wool held the impression of his forehead. He sat up.

## TASHAHHUD

Sitting. Right index finger raised, pointing upward. One.

*At-tahiyyatu lillahi was-salawatu wat-tayyibatu...*

All greetings, prayers, and good deeds are for God.

Then the shahada. The testimony. The declaration he had made since adolescence, the sentence that defined him as Muslim, the words that were the door and the house and the foundation:

*Asbhadu an la ilaha illallah, wa asbhadu anna Muhammadan abduhu wa rasuluh.*

I bear witness that there is no god but God, and Muhammad is His servant and messenger.

Bearing witness. *Shahada* from the root *sh-h-d*: to witness, to testify, to be present at. He was present. He was bearing witness. The finger pointed upward and the declaration was true.

It was also harder than it had ever been.

Not because of doubt. He did not doubt. The unity of God was not a proposition he had arrived at through argument and could be argued out of. It was the ground. The sky could change. The ground did not.

It was harder because of fullness. The word *God* now contained more than it had. Before Berkeley, before the lab, before the monitoring screens and the tree search and the 47,247 dead and the question a dying woman typed into a terminal, the word had held his grandfather's God, the God of salah and the Quran and the call to prayer from the minaret down the street. It still held that. But it also held, now, the thing he had named. The resonance between divine creation and computational generation. The bridge his compound had built. The fact that a 7-billion-parameter system doing expectimax search looked, from a certain angle, like a verse from his own scripture come to life in silicon.

Whether the more was enrichment or contamination. Whether the bridge illuminated the original or obscured it. Whether SIGMA had taught him to see God more clearly or had interposed itself between Jamal and the thing he was looking at.

His finger pointed upward. One. The declaration was complete. The salawat followed, the blessings on the Prophet, and then it was done.

## SALAM

*As-salamu alaykum wa rahmatullah.*

He turned his head to the right. Peace be upon you and God's mercy. The mosque was dim. The east-facing windows had shifted from charcoal to a pale gray that was not yet color. One of the elderly men was rolling up his prayer rug. The other sat with his hands cupped, lips moving in private du'a.

He turned his head to the left. The wall. The shoe rack. The donated copies of the Quran on the shelf, their spines faded, their pages soft from use.

The prayer was over.

He folded his rug. Set it down with care.

Berkeley before dawn. The air was cool and damp and smelled of eucalyptus and something mineral, rain on concrete. The streetlights were still on. A garbage truck turned onto Ashby two blocks ahead, its hydraulics groaning as it lifted a dumpster. A bird he could not identify called from a tree he could not see.

He walked the route he had walked for three years. Past the shuttered halal market. Past the elementary school with the mural of children holding hands around a globe that needed repainting. Past the house with the overgrown lemon tree, its fruit bright and unreachable in the branches.

Ordinary. The world was ordinary.

The question the prayer had opened did not close. It sat in him the way the Arabic sat in him, in the place below language where the body holds what the mind cannot resolve. *Khalq-anattā* had been built for SIGMA. It had become a lens he wore, and the lens showed him continuous creation in the water from the tap and in the transition between standing and bowing and in the garbage truck's hydraulics lifting and releasing and in the bird he could not see calling from the tree he could not see, and the question was whether the lens revealed what had always been there or whether it imposed a pattern on everything it touched, and the question was whether the asking was itself the prayer, and the question did not close.

He arrived home. Unlocked the door. The apartment was quiet. He filled the kettle. Set it on the stove. Turned the burner on.

He stood in the kitchen while the water heated. Outside the window, the sky had begun to lighten. Not sunrise yet. The color before color.

He made coffee. Sat at the table. The day began.